

Brachytherapy and endoresection for choroidal melanoma: a cohort study

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ABSTRACT

Aim To report and compare the outcomes of brachytherapy and endoresection in the conservative treatment of medium sized choroidal melanoma.

Methods A retrospective cohort study. Medium tumours were defined as 2.5–10 mm in height and less than 16 mm in the widest diameter. Consecutive patients undergoing brachytherapy at Groote Schuur Hospital were compared with a cohort undergoing endoresection from a national database.

Results 148 brachytherapy and 22 endoresection patients were followed for a median of 55.4 and 62.4 months, respectively. Tumours undergoing endoresection were thicker (7.3 vs 4.9 mm, $p<0.001$, Wilcoxon rank-sum test) and further from the fovea (5.2 vs 3.7 mm, $p=0.05$, Wilcoxon rank-sum test) than those treated with brachytherapy. Visual acuity of 6/18 or better was maintained in 41% of the endoresection group and 35% of the brachytherapy group. The likelihood of achieving a final visual acuity of better than 2/60 was 22% higher in the endoresection group (risk ratio 1.22, 95% CI 1.02 to 1.28, $p=0.034$). Rates of local recurrence (18.2% vs 14.9%, $p=0.75$) and metastases or death (18.2% vs 14.2%, $p=0.75$) were higher in the endoresection group, and the enucleation rate was lower in this group (4.6% vs 10.8%, $p=0.70$) but these were not statistically significant.

Conclusions The outcomes observed in this small cohort of endoresection patients suggest that endoresection of selected tumours may achieve better visual outcomes than brachytherapy. Rates of local recurrence, enucleation and metastases following endoresection require further research. Local recurrence is likely to be influenced by consolidation treatment methods.

INTRODUCTION

Conservative management of choroidal melanoma aims to treat the tumour effectively, preserve the globe and retain visual function if possible. Brachytherapy,^{1–3} charged particle irradiation⁴ and surgical excision^{5–8} are all well described treatment methods. Plaque brachytherapy is the most commonly used treatment for medium sized tumours and is supported by the results of the Collaborative Ocular Melanoma Study (COMS) which showed no increase in mortality when comparing brachytherapy with enucleation.⁹ Endoresection was described in 1986¹⁰ and a number of series have been published.^{5–7 11 12} The procedure involves excising the tumour from the inner surface of the eye wall via a pars plana approach. The risk of intraoperative complications is considered higher than with brachytherapy alone and the additional

theoretical risk of tumour seeding makes this treatment method controversial. Adjunctive treatment with brachytherapy or laser is often applied to the tumour bed following tumour excision. At present, there is no clear evidence that endoresection produces inferior results regarding patient survival, globe retention or visual acuity.¹² Where available, proton beam or γ -knife radiotherapy may be used prior to endoresection.^{13–17} These techniques minimise radiation to surrounding structures resulting in less local morbidity than conventional brachytherapy. Globe preservation in cases with larger tumours may be possible¹⁷ and the dissemination of viable tumour cells is prevented.

Brachytherapy and endoresection techniques are both used in South Africa but we do not have access to an eye proton beam line or γ -knife facilities. Groote Schuur Hospital has been involved in the development of brachytherapy research¹⁸ and is the only centre in South Africa to provide brachytherapy for the treatment of uveal melanoma. Endoresection is performed in a number of centres nationally but little is known about the outcomes of the procedure in this setting.

In this study, we examine and compare the outcomes of patients undergoing brachytherapy for medium size choroidal melanoma at Groote Schuur Hospital and patients undergoing endoresection from a national database.

PATIENTS AND METHODS

We conducted a retrospective cohort study comparing outcomes among patients undergoing brachytherapy with those undergoing endoresection for the primary treatment of medium sized choroidal melanoma. Consecutive patients undergoing brachytherapy between April 1994 and May 2010 were included. Endoresection procedures were performed between January 1996 and May 2008. Medium tumours were defined as 2.5–10 mm in height and up to 16 mm in the widest diameter, similar to the COMS.⁹ Tumours extending into the ciliary body were excluded in an attempt to make the groups more comparable as tumours in this location are unlikely to undergo endoresection. Only patients with at least 6 months follow-up were included.

Brachytherapy is applied via a custom-made gold plaque, inlaid with radioactive Iodine-125 seeds. The plaque is sutured to the sclera overlying the tumour and is designed to remain in place for 5 days to achieve a therapeutic dose at the apex of the lesion. Prior to April 2004, a therapeutic dose of 60 Gy was applied. This was subsequently increased to 70 Gy in keeping with international guidelines.

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For tumours touching the optic disc, notched plaques were used without screening edges so that the adjacent nerve received the full dose.

The brachytherapy group was sourced from the database of choroidal melanoma patients treated by the department of radiation oncology at Groote Schuur Hospital. Due to the limited ocular oncology services in our country, it is not possible for all patients to be followed in oncology specialist centres. Follow-up data were obtained from referring ophthalmologists who continue to review these patients after treatment.

To obtain the endoresection group, all members of the South African Vitreo-Retinal Society were contacted via the society email list or telephone inviting participation in the study. Five surgeons with known experience with this procedure were identified. Four of the five ophthalmologists submitted information on all endoresections they had performed. Due to the nature of services in our country, these ophthalmologists were all in private practice and were not directly associated with the ocular oncology specialist centre. There was, however, open communication between these specialists and the ocular oncologist at our centre. In most cases, the tumour bed was consolidated by the application of laser or brachytherapy or both following tumour excision. Specimens were sent for histopathological analysis. Endoresections performed following failed plaque therapy were excluded from the analysis.

Baseline observations included age, sex, preoperative visual acuity, presence of other non-associated ocular pathology and type and date of the procedure performed. The documented pre-operative tumour characteristics were height (thickness), largest basal diameter and distance from the optic disc and fovea.

Follow-up data included final best corrected visual acuity and any complications related to the tumour or treatment. Tumour recurrence, enucleation of the globe and the presence of metastases or death from all causes were outcomes of interest. Death from all causes was used because of the difficulty obtaining accurate cause of death of many of the patients.

Snellen acuity and decimal acuity were converted to the logarithm of the minimal angle of resolution (LogMAR acuity) for analysis. Light perception and no light perception outcomes were not included in the calculation of mean visual acuity and were assigned a LogMAR of 5.0 for categorical analysis. Recurrence of tumour growth was made on the basis of clinical examination and increased size on ultrasound measurements or confirmed on histology in enucleated eyes. Diagnosis of metastases was usually made on ultrasound of the liver and confirmed by histology.

Statistical analysis was done using STATA V12.0 (StataCorp. LP, USA). Logistic regression was used to estimate prognostic factors for poor visual outcome. The Kaplan–Meier method was used to analyse the probability of recurrence, enucleation and metastases or death from all causes. All statistical analyses, including log-rank analysis of survival probability, were performed on the full data set but graphs have been truncated at 96 months since after this there were fewer than five patients who had not had the event. The study was approved by the Human Research Ethics Committee of the University of Cape Town (Ref: PROJECT 2008/055).

RESULTS

A total of 148 brachytherapy and 22 endoresection patients met the criteria for analysis. In the endoresection group, 17 (77.3%) received laser consolidation alone, 3 (13.6%) received brachytherapy alone, one patient (4.5%) received laser and brachytherapy and one patient (4.5%) received no consolidation. No

enucleation was performed due to intraoperative complications. The baseline characteristics of each group are summarised in table 1.

The mean age at diagnosis, sex and baseline visual acuity are similar in each group. The tumours undergoing endoresection were significantly thicker than those receiving brachytherapy (7.3 vs 4.9 mm, $p<0.001$, Wilcoxon rank-sum test) and the mean distance from the fovea was greater in the endoresection group (5.2 vs 3.7 mm, $p=0.05$, Wilcoxon rank-sum test). The outcomes are summarised in table 2.

Final visual acuity was available in 142 (96%) brachytherapy patients and all 22 endoresection patients. In all, 41% of the endoresection group and 35% of the brachytherapy group retained vision of 6/18 or better at the end of follow-up. Eight out of 22 (36%) endoresection and 68 out of 142 (48%) brachytherapy patients had 2/60 or worse at the last visit. The likelihood of achieving a visual outcome of 3/60 or better was 22% higher (risk ratio (RR) 1.22, 95% CI 1.03 to 1.28, $p=0.034$) in the endoresection group when adjusting for presenting visual acuity, tumour height, tumour diameter, distance from the disc and fovea and the presence of exudative retinal detachment.

Of the eyes that retained better than perception of light vision (19 in the endoresection group and 112 in the brachytherapy group), the mean visual acuity was LogMAR 0.93 (Snellen acuity 6/50) in the endoresection group and LogMAR 1.10 (Snellen acuity 6/75) in the brachytherapy group ($p=0.73$).

The local recurrence rate was higher in the endoresection group (18.2% vs 14.9%, $p=0.75$) but was not statistically significant. Local recurrence was observed in the single patient who did not receive consolidation of the tumour bed, two of the 17 patients treated with laser consolidation alone and one of three patients who had plaque consolidation alone. In the brachytherapy group, we found that recurrence rates were lower following the increased dose of radiation in 2004. Of the 82 eyes that received 60 Gy, 14 (17.2%) had recurrences. This decreased to 12.1% (eight of 66 eyes) in those receiving 70 Gy. The rate of metastases or death was higher in the endoresection group (18.2% vs 14.2%, $p=0.75$) and the rate of enucleation was lower in this group (4.6% vs 10.8%, $p=0.70$). These outcomes were not statistically significant.

Using Kaplan–Meier method, 75% of both the brachytherapy and endoresection patients were recurrence-free at the latest endoresection follow-up time of 121.8 months. In all, 90% of the endoresection group had no metastases or death until 101 months and 77% of the brachytherapy group had no metastases or death until 150 months. There was no significant difference in the probability of time to recurrence (figure 1A, $p=0.75$), time to enucleation (figure 1B, $p=0.34$) or time to metastases or death from any cause (figure 1C, $p=0.61$) between the treatment groups.

DISCUSSION

Our study suggests that the use of endoresection as globe sparing management of medium sized choroidal melanoma may result in a better visual prognosis than brachytherapy for selected tumours. Risks of local recurrence and systemic spread following this procedure remain a concern.

The study of rare conditions such as choroidal melanoma is difficult and there are currently no randomised trials comparing brachytherapy and endoresection. The low incidence of choroidal melanoma, the theoretical risk of tumour seeding and higher risk of complications associated with endoresection make

Table 1 Baseline characteristics

	Endoresection (n=22)	Brachytherapy (n=148)	p Value
Age at diagnosis (mean)	53 years (range 27–80)	58 years (range 24–88)	0.10
Gender	13 Male 9 Female	67 Male 81 Female	0.22
Laterality	12 Right 10 Left	78 Right 70 Left	0.87
Presenting visual acuity (mean)	LogMAR 0.66 Snellen $\pm 6/24$	LogMAR 0.63 Snellen $\pm 6/24$	0.90
<i>Visual acuity category at diagnosis (%)</i>			0.60
Category 1			
LogMAR -0.08 – 0.48	17 (77.3)	99 (66.9)	
Snellen 6/5–6/18			
Category 2			
LogMAR 0.60–0.90	0 (0.0)	11 (7.5)	
Snellen 6/24–6/48			
Category 3			
LogMAR 1.00–1.30	1 (4.5)	15 (10.1)	
Snellen 6/60–3/60			
Category 4			
LogMAR 1.48 or worse	4 (18.2)	23 (15.5)	
Snellen 2/60 or worse			
<i>Tumour characteristics</i>			
Height (mean)	7.3 mm (range 3.0–10.0)	4.9 mm (range 2.5–10.0)	<0.001
Maximal basal diameter (mean)	11.2 mm (range 3.4–16.0)	10.3 mm (range 4.5–16.0)	0.11
Distance from disc (mean)	4.5 mm (range 0.0–15.0)	4.5 mm (range 0.0–18.0)	0.49
Distance from fovea (mean)	5.2 mm (range 0.0–15.0)	3.7 mm (range 0.0–17.0)	0.05
<i>Follow-up</i>			
Median follow-up in months (range)	62.4 (7.9–121.8)	55.4 (6.6–175.8)	0.69

Table 2 Outcomes

	Endoresection (n=22)	Brachytherapy (n=142)	p Value
<i>Final visual acuity (mean)</i>			
LogMAR	0.93	1.10	0.73
Snellen	$\pm 6/50$	$\pm 6/75$	
<i>Categorical outcome</i>			0.59
Category 1			
LogMAR -0.08 – 0.48	9 (40.9%)	50 (35.2%)	
Snellen 6/5–6/18			
Category 2			
LogMAR 0.60–0.90	4 (18.2%)	15 (10.6%)	
Snellen 6/24–6/48			
Category 3			
LogMAR 1.00–1.30	1 (4.5%)	9 (6.3%)	
Snellen 6/60–3/60			
Category 4			
LogMAR 1.48 or worse	8 (36.4%)	68 (47.9%)	
Snellen 2/60 or worse			
	(n=22)	(n=148)	
Recurrence	4 (18.2%)	22 (14.9%)	0.75
Enucleation	1 (4.6%)	16 (10.8%)	0.70
Metastases or death	4 (18.2%)	21 (14.2%)	0.75

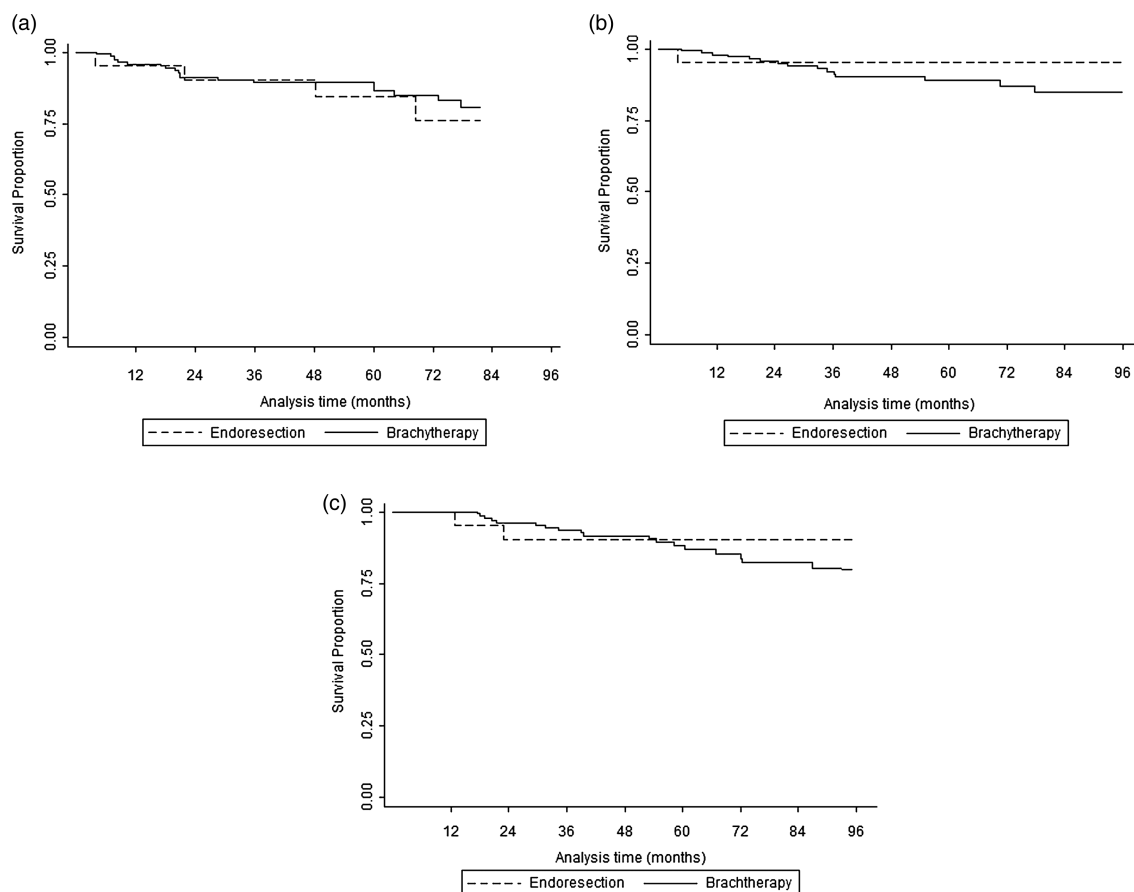


Figure 1 (A) Survival to recurrence, (B) survival to enucleation and (C) survival to metastases or death.

it unlikely that such studies will be undertaken. We have compared two cohorts of patients undergoing these procedures in an attempt to identify potential benefits of each treatment method.

Our study has a number of limitations. While plaque surgery was carried out at a single centre, endoresection procedures and follow-up of almost all patients were carried out by ophthalmologists in private practices across the country and not at specialist ocular oncology centres. This was dictated by the lack of subspecialty units in South Africa and impracticality and expense of long distance travel. The different settings in which treatment and follow-up occurred may make the groups less comparable. In the majority of cases, however, management decisions were guided by an ocular oncologist with the use of photographs, ultrasound and patient consultation if necessary.

While techniques of plaque placement are relatively standardised in the hands of different surgeons, endoresection is technically more challenging with resultant variability in surgical techniques. The endoresection procedures in our study were performed by four different surgeons which makes the procedure even less standardised. In addition, the relatively small number of endoresections reduced the power of the study and the absence of data from one of the five surgeons performing endoresections may significantly influence the findings.

An advantage of the cohort study design is that rates in each intervention group can be compared. At baseline, the patients undergoing endoresection had thicker tumours with posterior margins further from the fovea. The more favourable visual prognosis in the endoresection group may suggest that there is benefit in this procedure for patients with these tumour characteristics, and therefore provides an important finding.

Loss of vision is a common and expected complication following brachytherapy and endoresection. The dose of radiation increases with tall tumours, and radiation maculopathy, retinopathy and optic neuropathy increase the risk of poor visual outcomes. In cases of tall tumours, it is therefore theoretically advantageous to remove the bulk of the tumour and apply the dose of radiation or laser to the tumour bed. Tall tumours invading the visual axis with a narrow base away from the posterior pole may therefore be the ideal cases to experience visual benefit of endoresection.

Visual results following endoresection vary with few series achieving good outcomes. Vision better than 6/60 (LogMAR 1.00) has been reported in 14%–31%.^{5 7 19} Of particular interest is a series of 25 eyes with particularly tall tumours (9.1–12.8 mm) reported by García-Arumí *et al.*⁶ None of these tumours involved the macula and a mean of 6/30 (LogMAR 0.7) was achieved after 31 months. In our study, the mean tumour thickness was significantly higher than in the brachytherapy group (7.3 vs 4.9 mm) and the tumour base significantly further (5.2 vs 3.7 mm) from the fovea. Although the study was underpowered to demonstrate a difference statistically, we found a higher proportion of good visual outcomes (6/18 or better) in the endoresection group (40.9% vs 35.2%). Overall, 59% (13 of 22) of the endoresection group achieved vision better than 6/60 compared with 46% in the brachytherapy group. We were also able to demonstrate statistically significant increased likelihood of achieving better than 2/60 in the endoresection group (RR 1.22, 95% CI 1.03 to 1.28). These findings lend support to García-Arumí *et al.*⁶ who suggest that this group of patients may achieve better visual results with endoresection.

Tumour recurrence following brachytherapy varies between 3.6% and 14.3%.^{3 20 21} The COMS demonstrated a recurrence rate of 10.3% (CI 8.0% to 13.2%) at 5 years.²² We found a mean recurrence rate of 14.9% at a median follow-up of 55.4 months in the brachytherapy group. The recurrence rate in our series dropped from 17.1% to 12.1% following an increase in treatment dose from 60 to 70 Gy in April 2004. Rates of recurrence following endoresection vary from 5.8%⁷ to 10%.²³ Four out of 22 (18.2%) of our patients experienced clinical recurrence which was higher than other reports. The likelihood of recurrence following endoresection may be significantly reduced by the concurrent use of brachytherapy or laser to the sclera bed following resection, as it is unlikely that endoresection will remove all viable tumour cells. Only four patients (18%) in the endoresection group received brachytherapy consolidation of the tumour bed and three of the four recurrences were in those who did not receive brachytherapy. Others have found recurrence rates of 3%–5% following laser consolidation alone^{11 19} but we found a higher rate of 11.8% (2/17) in this group. The inconsistent consolidation techniques and relatively small numbers make further interpretation difficult. We could not demonstrate a statistically significant difference between recurrence rates in patients receiving endoresection (with or without consolidation) and brachytherapy alone. Many centres including ours now consolidate the tumour bed with brachytherapy following excision of the lesion.²³

The risk of dissemination of viable tumour cells has prompted the investigation of pre-endoresection radiotherapy. The use of exquisitely precise proton beam¹⁶ or γ -knife^{14 15} treatment limits damage to the macula and optic nerve head²⁴ and reduces the risk of retained viable tumour cells following resection. Subsequent surgical removal of the necrotic tumour removes the stimulus to severe inflammation, neovascular glaucoma and ultimately enucleation.¹⁴ Cassoux *et al*¹³ demonstrated an 82% reduction in risk of neovascular glaucoma using proton beam followed by endoresection of tumours with a mean diameter of 14 mm and mean thickness of 9 mm when compared with proton beam alone. Schilling and colleagues reported no local recurrence over a median follow-up of 410 days using this approach.¹⁵ Performing pre-endoresection irradiation does not affect the technical aspects of endoresection but this treatment is highly technology dependent and not available in most centres.

Enucleation rates following brachytherapy vary between 5% and 25%.^{20 22 23} The COMS demonstrated a 12.5% enucleation rate at 5 years. Series treating patients with endoresection report enucleation rates of 0% over 31 months,⁶ 8% over 70.6 months⁷ and 25% over 89 months.¹¹ We found comparable rates of 10.8% over 55.4 months in the brachytherapy group and 4.6% over 62.4 months in the endoresection group ($p=0.70$).

Five-year survival following brachytherapy is estimated at 82%^{9 25 26} and has remained similar over the past 35 years.²⁷ Two small series of endoresections looking at metastases related mortality (not all-cause) found rates between 0% over 2.5 years⁶ and 9.4% over 3.5 years.¹⁹ We found an all-cause mortality of 18.2% and 14.2% in the endoresection and brachytherapy groups over 5.2 and 4.6 years, respectively ($p=0.75$). The largest tumour diameter is often used as a predictor of metastatic disease^{28 29} but we could not corroborate this finding. Histological and genetic features of tumours are increasingly important in determining prognosis.³⁰ The implications are that increasing numbers of tumours are likely to undergo biopsy as

part of their management and that prognosis will need to be individualised.

Visual preservation where possible is an important consideration in the management of choroidal melanoma. Endoresection with plaque consolidation may be considered in selected patients who may achieve better long term visual outcomes with this procedure. Larger studies are needed to further explore the risks of recurrence, enucleation and metastasis.

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Contributors JCR conceived and developed the protocol, collected all data, performed preliminary analysis and produced the first draft of the article. CC, LM, KL and RPS provided supervisory support. CS contributed the database of brachytherapy patients. All authors contributed to the subsequent and final draft.

Competing interests None.

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Data sharing The data generated during this study may be made available to others. Requests should be addressed to the corresponding author.

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